

30 Days SQL Micro Course Certificate Assignment

1. Total Number of Patient

Find the total number of patients across all hospitals.

Answer :-

```
SELECT SUM(patients_count) AS Total_patients  
FROM hospital;
```

2. Average Number of Doctors per Hospital

Retrieve the average count of doctors available in each hospital.

Answer :-

```
SELECT hospital_name, AVG(doctors_count) AS Average_doctors  
FROM hospital  
GROUP BY hospital_name;
```

3. Top 3 Departments with the Highest Number of Patients

Find the top 3 hospital departments that have the highest number of patients.

Answer :-

```
SELECT department, SUM(patients_count) AS Total_patients  
FROM hospital  
GROUP BY department  
ORDER BY total_patients DESC LIMIT 3;
```

4. Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.

Answer :-

```
SELECT hospital_name, SUM(medical_expenses) AS total_medical_expenses
```

```
FROM hospital
GROUP BY hospital_name
ORDER BY total_expenses DESC LIMIT 1;
```

5. Daily Average Medical Expenses

Calculate the average medical expenses per day for each hospital.

Answer :-

```
SELECT hospital_name, AVG(medical_expenses / (discharge_date-admission_date)) AS
avg_expens_per_day
FROM hospital
GROUP BY hospital_name;
```

6. Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge Date and Admission Date.

Answer :-

```
SELECT *, (discharge_date - admission_date) AS stay_days
FROM hospital
ORDER BY stay_days DESC
LIMIT 1;
```

7. Total Patients Treated Per City

Count the total number of patients treated in each city.

Answer :-

```
SELECT location, SUM(patients_count) AS total_patients
FROM hospital
GROUP BY location;
```

8. Average Length of Stay Per Department

Calculate the average number of days patients spend in each department.

Answer :-

```
SELECT department, AVG(discharge_date - admission_date) AS avg_stay_days
FROM hospital
GROUP BY department;
```

9. Identify the Department with the Lowest Number of Patients

Find the department with the least number of patients

Answer :-

```
SELECT department, SUM(patients_count) AS total_patients
FROM hospital
GROUP BY department
ORDER BY total_patients ASC
LIMIT 1;
```

10. Monthly Medical Expenses Report

Group the data by month and calculate the total medical expenses for each month.

Answer :-

```
SELECT EXTRACT(MONTH FROM admission_date) AS month, SUM(medical_expenses)
AS total_medical_expenses
FROM hospital
GROUP BY month
ORDER BY month;
```
